

Modelling of Large-Scale Circulation in Atmosphere and Ocean
(MO8004)

Wind Driven Ocean Gyres Sverdrup and Munk Solutions

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1 Introduction

Due to the Hadley cells together with the Coriolis effect, tradewinds and westerlies dominate the winds over the North Atlantic [Hartmann, 2016]. These winds drive a converging Ekman transport, which is stronger in the North due to the lateral dependence on the Coriolis force [Ekman, 1905]. This imbalance forces a net southward flow and is known as the Sverdrup circulation [Sverdrup, 1947], and dominates most extratropical oceans.

Stommel [1948] further developed the solution, by introducing drag which gave return flow on the western boundary. The western boundary current was further developed by Munk [1950], who assumed a viscous fluid. In this report, the Sverdrup and Munk solutions will be examined by using a set of linear shallow water equations with a Fortran code. This will give insight into the two ocean gyre theories and how a numerical model handles them respectively.

2 Theory

One can approximate the wind stress in the North Atlantic with

$$\vec{\tau} = \tau_x = \rho_{\text{air}} C_D U_{10}^2 \sin\left(\frac{\pi y}{L_y}\right), \quad (1)$$

where ρ_{air} is the air density, C_D is the drag coefficient, U_{10} is the ten meter wind speed, and L_y is the lateral extent of the basin. The sinusoidal dependence represents easterly trade winds in the southern part of the North Atlantic and westerlies in the northern part.

2.1 Sverdrup Theory

In order to study Sverdrup circulation one needs to make assumptions: One needs a set of linear and stationary shallow water equations, a meridionally varying ocean surface stress, the beta plane approximation as

$$f \approx f_0 + \beta y, \quad (2)$$

an inviscid fluid and a flat bottom. One also needs the ocean to be laterally bounded, for example by two continents on either side [Vallis, 2006]. One also assumes a barotropic ocean. This is however not completely true, since an Ekman layer with Ekman transport is assumed, but since the size of this layer is much smaller than the domain, it is sufficient to say that the ocean is barotropic. The barotropic streamfunction is defined as

$$\frac{\partial \Psi}{\partial x} = vH, \quad \frac{\partial \Psi}{\partial y} = -uH, \quad (3)$$

where H is the mean depth of the ocean. Using this one obtains the Sverdrup balance as

$$\beta vH = \frac{1}{\rho} \nabla \times \vec{\tau}. \quad (4)$$

One can thus derive the Sverdrup solution, by assuming $\Psi(x_{\text{East}}, y) = 0$, as

$$\int_x^{x_{\text{East}}} \frac{\partial \Psi}{\partial x} dx = \int_x^{x_{\text{East}}} vH dx = \Psi(x_{\text{East}}, y) - \Psi(x, y) \quad (5)$$

$$\implies \Psi(x, y) = -\frac{1}{\rho\beta} \int_x^{x_{\text{East}}} (\nabla \times \vec{\tau}) dx, \quad (6)$$

The Sverdrup solution thus gives a net southward flow in the center of the domain. The solution is a "parabolic-like" structure along the longitudinal direction, with the decreasing values towards the East. The western boundary will however not have return flow, and the streamlines will not be closed.

2.2 Munk Theory

In Munk theory a meridional harmonic Newtonian viscosity to the linear shallow water equations is added. The viscosity is diffusive, which agrees with the fact that the ocean is turbulent and thus has an effective viscosity [Vallis, 2006]. The reason for only assuming meridional viscosity, is that this is the only net velocity in the Sverdrup solution. By assuming that the solution is bounded by the domain, $\Psi(x_{\text{East}}) = \Psi(x_{\text{West}}) = 0$, one obtains

$$A\nabla^4\Psi - \beta\frac{\partial\Psi}{\partial x} = \frac{1}{\rho}\nabla\times\vec{\tau} \implies \quad (7)$$

$$\Psi(x, y) = \Psi_{\text{Sverdrup}}(x, y) \left(1 - e^{-\frac{x-x_{\text{West}}}{2\delta_M}} \left[\cos\left(\sqrt{3}\frac{x-x_{\text{West}}}{2\delta_M}\right) + \frac{1}{\sqrt{3}}\sin\left(\sqrt{3}\frac{x-x_{\text{West}}}{2\delta_M}\right) \right] \right), \quad (8)$$

where δ_M is the Munk layer defined as

$$\delta_M \equiv \left(\frac{A}{\beta}\right)^{\frac{1}{3}}. \quad (9)$$

The Munk layer is the distance from the boundary, where the new frictional Munk term dominates over the Sverdrup term. One thus assumes that the interior should behave as the Sverdrup solution. Due to its higher order the Munk theory will exhibit another shape. This will yield closed streamlines at the western boundary, with a western boundary current at the distance δ_M from the western boundary, since the Munk term opposes the sign of the Sverdrup term. However the Sverdrup interior solution will remain.

3 Methodology

A model written in Fortran was used that implemented the linear shallow water equations, with a stress described by Equation 1. The code was initialized at rest and used an Euler step for the first step, then a leapfrog scheme on a Arakawa C-grid as presented by Döös et al. [2022]. The code used a time-step derived from an appropriate Courant number. The code ran for at 80 days, and was saved every 12th hour. The code can be observed in Appendix A.

The code used a $H = 3000\text{ m}$, $f_0 = 7.27 \times 10^{-5}\text{ s}^{-1}$ and $\beta = 1.98 \times 10^{-11}\text{ m}^{-1}\text{ s}^{-1}$. The model used a domain with solid boundaries of the size $L_x = L_y = 7000\text{ km}$ to represent the Atlantic Ocean (twice as wide for the Pacific Ocean) with $\Delta x = \Delta y = 35\text{ km}$. Varying the stress was done by using $U_{10} = [4\text{ m s}^{-1}, 5\text{ m s}^{-1}, 6\text{ m s}^{-1}]$, and varying the viscosity was done by $A = [3 \times 10^4\text{ m}^2\text{ s}^{-1}, 1 \times 10^5\text{ m}^2\text{ s}^{-1}, 5 \times 10^5\text{ m}^2\text{ s}^{-1}]$. The harmonic viscosity in the boundary points assumed zero flow outside the domain, leading to energy dissipation.

4 Results

4.1 Sverdrup circulation

The Sverdrup solution gave Figure 1–2. One can in Figure 1 see the result of the f -plane, which does not resemble the beta Sverdrup theory. Here the solution is symmetric, since there is no latitudinal asymmetry caused by the β -term. Furthermore, the solution is growing with time, due to the continuous input of wind energy. This is a growing geostrophic solution to the applied wind stress, and thus symmetric in the domain. However, an gyre can still be seen due to the consequence of the y -dependence on the wind stress.

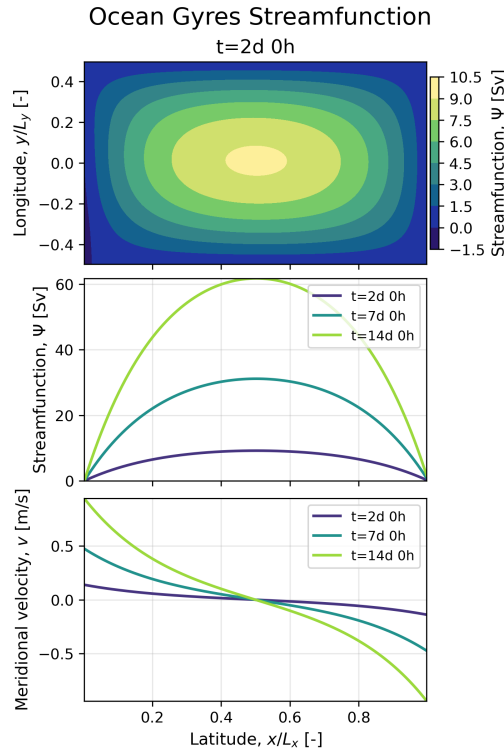


Figure 1: Streamfunction for the no-viscosity f -plane, for different times.

The introduction of the β -term gave Figure 2. One can observe Sverdrup like solution compared to Figure 1, due to the asymmetric beta effect (stronger f northward and weaker southward) pushing the solution towards the western boundary. The mirror-symmetry is thus broken and a better Sverdrup solution is obtained.

By increasing the wind stress one can observe linearly increases in the flux of the domain. This is reasonable since the wind stress is the forcing that drives this circulation. Thus, more forcing yields larger fluxes. Examining the slope, one sees larger southward velocities in the interior. Overall, one can see good agreement for the time-averaged result and the theory. Away from the western boundary the model and the theory align very good, but close to the western boundary they align very bad.

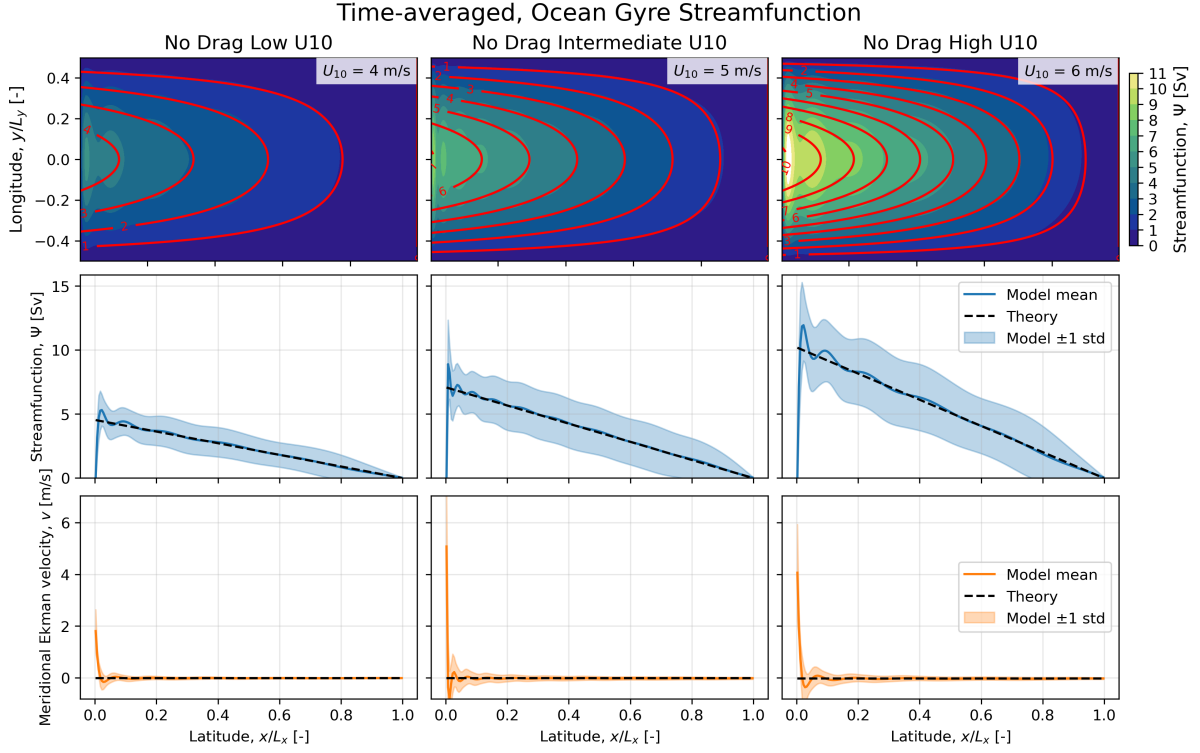


Figure 2: Comparison of streamfunction for different wind stresses. The contours show the model result, while the solid red lines showcase the Sverdrup theory. The cross-sections of the streamfunction is at the center of the domain. The meridional wind velocity has been transformed into an Ekman-layer velocity, by assuming an Ekman depth of $\delta_{EK} = 50$ m. The meridional velocity is thus the corresponding mass flux as if the domain is at rest and only the Ekman layer is moving. The time averaging was done between two days and eighty days.

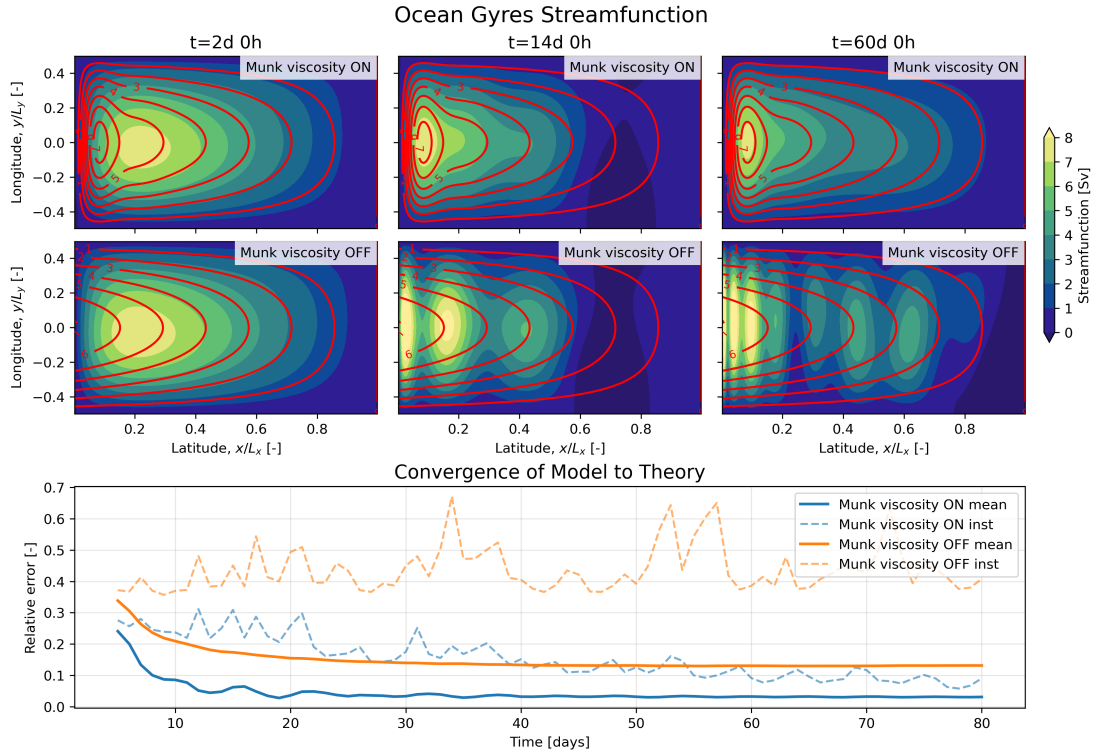


Figure 3: This figure shows the effect of the Munk friction. The red lines are the theoretical Sverdrup and Munk solutions respectively.

4.2 Munk's theory

The difference of including Munk viscosity gave Figure 3. One can observe that the Munk theoretical results now include the western boundary current, in both the numerical and theoretical solution. Thus, the numerical solutions agrees better with the theoretical, compared to the Sverdrup solution. The largest distinction is the steady state convergence of the Munk model, which does not occur for the Sverdrup model. Observing the instantaneous relative error one can see a convergence towards lower errors for the Munk solution, while the same cannot be said for the Sverdrup solution. One thus needs to time average the Sverdrup result to obtain good values. One can observe that the time-averaged Sverdrup solution gives similar accuracy to the instantaneous Munk solution. However, to make a fair comparison between the two models, the Munk solution was also time averaged.

Munk theory gave Figure 4. Increasing the viscosity coefficient A can be seen to do two major things: broadening the western boundary current and thus also lowering the northward velocity. The location of the western boundary current can be observed to be precisely at the Munk layer δ_M . Compared to the Sverdrup solution one observes better result in Figure 4, especially for the western boundary current.

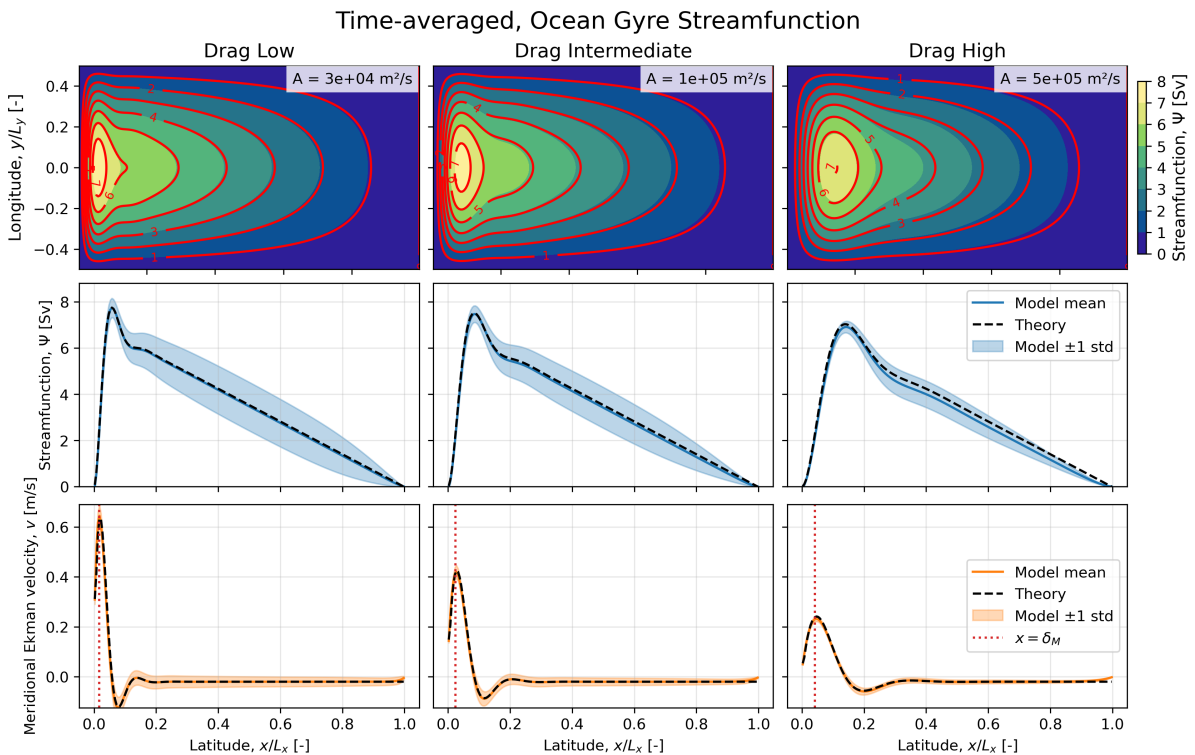


Figure 4: Streamfunction for different values of the viscosity coefficient A . The contours show the model result, while the solid red lines showcases the theory. The resolution of the Munk depth is $\frac{\delta_M}{\Delta x} = [3.8, 4.9, 8.4]$, from left to right in the figure. The center domain zonal cross-sections show solid lines for the numerical model, while dashed lines represent Munk theory. The meridional wind velocity has again been transformed into an Ekman-layer velocity (as explained in Figure 2). The time-averaging was done between two days and eighty days.

In Figure 5 one can observe the effect of increasing the domain width, while keeping Δx constant. One can observe a larger flux for the North Pacific compared to the North Atlantic, although the meridional velocity is exactly the same in the two oceans (as seen from the slope of the streamfunctions). This also provides the larger domain with a faster western boundary current.

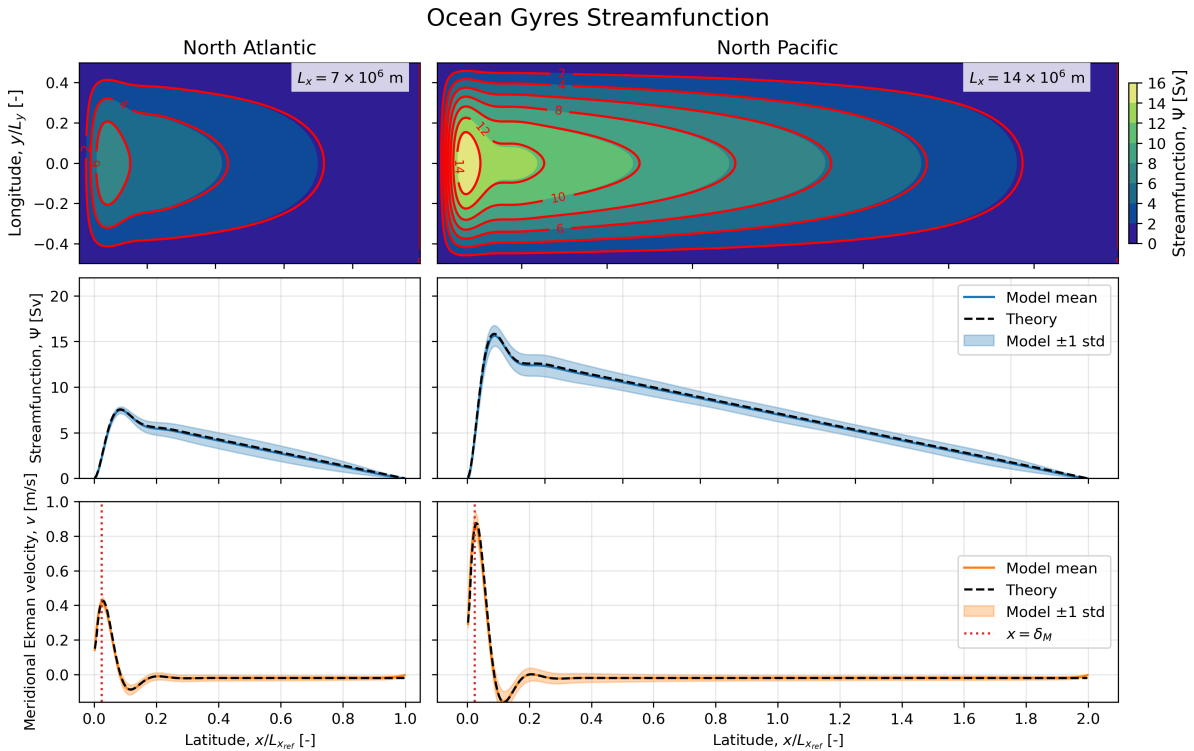


Figure 5: Streamfunction for the viscosity configuration with varying domain size. The contours show the model result, while the solid red lines showcases the Munk theory. In the center domain cross-section, solid lines show the numerical model results, while dashed lines show theory. The meridional wind velocity has been transformed into an Ekman-layer velocity. The time averaging was done between two days and eighty days.

5 Discussion

The reason for the western intensification can be explained by the phase movement of Rossby waves always being westward. Thus, larger fluxes appear on the western half. In the Munk theory, the imposed harmonic viscosity ensured that these Rossby waves died down, ensuring a stable solution. However, the lack of diffusion in the Sverdrup solution forced continuous Rossby wave propagation, which explains the non-converging result. One can equally explain this theoretically by a mass accumulation along the western boundary creating a high which propagates like a Rossby wave. This explains the clear limitation of the solution not reaching steady state. However, since the Rossby wave propagation was superimposed on the solution, time averaging resulted in good results.

The Sverdrup non-convergence can be explained through the lack of energy dissipation. Through continuous energy input, propagation of Rossby waves was continuously fed. This is the same mechanism that forced the f -plane solution to grow continuously. Thus one needs some kind of dissipation to balance the continuous energy input. In the real world this is done through small-scale dissipation, connections to other oceans and other dynamical properties.

Another problem of the Sverdrup solution is the lack of mass convergence in the theory. Since the theory is first order, only one boundary condition could be imposed, which cannot allow return flow (although this was forced numerically through mass conservation). The Munk solution is higher order, allowing multiple boundary conditions and a return flow. The return flow can be explained through the frictional effect opposing the mean southward flow, by creating a boundary layer close to the western boundary. Since frictional effects act in the opposite direction, a strong northern current is created.

One can furthermore criticize the Munk model for artificial viscosity. If the Stommel drag is the friction caused by bottom drag, the Munk viscosity should be a boundary effect, and thus only be seen at the boundaries. However, although the viscosity was implemented throughout the entire domain, the Munk

layer ensured that this effect was only present near the boundary. Furthermore, since the viscosity presents reasonable results, one can argue that the model is of good use. However, in most realistic models one often uses the combined Munk–Stommel viscosity to obtain benefits of both models [Vallis, 2006].

One can always obtain better results by increasing the grid size in the domain, for example to resolve the western part of the boundary current in the Munk model. Since the lowest viscosity only had a resolution of $\frac{\delta_M}{\Delta x} = 3.8$, one can observe a non zero value at the western boundary. Thus increasing Δx one could obtain a higher resolution and $\Psi(x_{\text{West}}) \rightarrow 0$. However, according to Steele [2008], the speed of the Gulf Stream is around 2.5 m s^{-1} . Comparing this value with Figure 2–4, the obtained results in these schemes are thus on the right order of magnitude compared to the real world. Thus, even though this article presented a simplified model, reasonable values were obtained. Furthermore, the theoretical error remained below 15% for both models, showcasing working numerical models.

6 Conclusion

In conclusion, both models have their strengths and weaknesses. Sverdrup theory is a good intuitive way to visualize the flow in the mid ocean, while Munk theory is good if one also wants to resolve the western boundary current. Three important conclusions are:

- Both numerical models showed western boundary intensification, explained by Rossby wave propagation. However these Rossby waves must be balanced by some kind of friction to die with time and not cause non-converging behaviour.
- Although both numerical models showed return flow, only Munk theory showed a western boundary current, explained by a frictional boundary layer.
- A larger domain gave the same interior meridional velocity, which resulted in a stronger western boundary current due to the larger flux of the larger ocean.

One can summarize the two models by stating that although the Sverdrup theory is more intuitive from a physical perspective, the artificial Munk theory presented a more realistic solution. However, a good physical-oceanographer would then know when to use one model or the other, in order to explain different aspects of the wind driven ocean gyres.

References

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A The Source Code

The source code for this lab can be found on GitHub:

`https://github.com/FredrikBergelv/
Modelling-of-Large-Scale-Circulation-in-Atmosphere-and-Ocean`